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$y = m_1 x_1 + m_2 x_2 + b$

where:

- Initial $m = [-1, 2]$
- Initial $b = [1, 1]$
- Given data points:

$x = (1,3), (4,10)$

$y = (5, 6)$

$$\begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 6 \end{bmatrix}$$

so: $m1 = -1$ $m2 = 2$ $b = 1$

$a = 0.01$

$$X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \quad X^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$$

Iteration 1:

1. Predict.

$$\begin{bmatrix} 1\cdot(-1) + 3(2) \\ 4\cdot(-1) \quad 10(2) \end{bmatrix} = \begin{bmatrix} -1+6 \\ -4+20 \end{bmatrix} = \begin{bmatrix} 5 \\ 16 \end{bmatrix}$$

using $y = mx + b$

$$\hat{y} = \begin{bmatrix} 5+1 \\ 16+1 \end{bmatrix} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$$

2. Error. ($e = \hat{y} - y$, then $mse = (e_1^2 + e_2^2) \div 2$

$$\begin{bmatrix} 6-5 \\ 17-6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix} \quad \text{so } J = (1^2 + 11^2) \div 2$$

$$1 + 121 = \frac{122}{2} = 61$$

3. Slopes. $\frac{2}{n} = \frac{2}{2} = 1$

$X^T \cdot e$

$$\begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 1\cdot 1 + 4\cdot 11 \\ 3\cdot 1 + 10\cdot 11 \end{bmatrix}$$

$$= \begin{bmatrix} 1+44 \\ 3+110 \end{bmatrix} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$\frac{\partial J}{\partial m} = 1 \times (45, 113) = 45, 113$

$\frac{\partial J}{\partial b} = 1 \times (1+11) = 12$

$1 + 11 = 12$

4.

$m \leftarrow m - a \cdot \partial J / \partial m$

$b \leftarrow b - a \cdot \partial J / \partial b$

$m_1 = -1 - (0.01 \times 45)$

$-1 - 0.45 = -1.45$

$m_2 = 2 - (0.01 \times 113)$

$2 - 1.13 = 0.87$

$b = 1 - (0.01 \times 12)$

$1 - 0.12 = 0.88$

so $m_1 = -1.45$

$m_2 = 0.87$

$b = 0.88$

Iteration 2 : Belyse Intwaza

from Iteration 1:

$m_1 = -1.45$ $m_2 = 0.87$ $b = 0.88$

$\alpha = 0.01$ $n = 2$

$X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$ $X^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$ $m = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$ $y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$

Chain rule: $J = 1/n \sum (\hat{y} - y)^2$ $Y = Xm + b$

$dJ/dm = dJ/d\hat{y} \cdot d\hat{y}/dm$

$= 2/n (\hat{y} - y) \cdot X$

$= 2/n \cdot X^T (\hat{y} - y)$

$dJ/db = dJ/d\hat{y} \cdot d\hat{y}/db$

$= 2/n (\hat{y} - y) \cdot 1$

$= 2/n \cdot \sum (\hat{y} - y)$

1. Predict $(\hat{y} = Xm + b)$

$Xm = \begin{bmatrix} 1(-1.45) + 3(0.87) \\ 4(-1.45) + 10(0.87) \end{bmatrix}$

$= \begin{bmatrix} -1.45 + 2.61 \\ -5.80 + 8.70 \end{bmatrix} = \begin{bmatrix} 1.16 \\ 2.90 \end{bmatrix}$

$\hat{y} = \begin{bmatrix} 1.16 + 0.88 \\ 2.90 + 0.88 \end{bmatrix} = \begin{bmatrix} 2.04 \\ 3.78 \end{bmatrix}$

2. Error $(e = \hat{y} - Y$, then $J = (e1^2 + e2^2) \div 2)$

$e = \begin{bmatrix} 2.04 - 5 \\ 3.78 - 6 \end{bmatrix} = \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix}$

$J = (-2.96)^2 + (2.22)^2 \div 2$

$= (8.7616 + 4.9284) \div 2$

$= 13.69 \div 2 = 6.845$

$= 6.845$

3. Slopes $(2/n = 2/2 = 1)$

$X^T e = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix}$

$= \begin{bmatrix} 1(-2.96) + 4(2.22) \\ 3(-2.96) + 10(-2.22) \end{bmatrix}$

$= \begin{bmatrix} -2.96 - 8.88 \\ -8.88 - 22.20 \end{bmatrix} = \begin{bmatrix} -11.84 \\ -31.80 \end{bmatrix}$

$dJ/dm = 1 \times (-11.84, -31.08)$

$= (-11.84, -31.08)$

$dJ/db = 1 \times (2.96 + -2.22) = -5.18$

4. Update $(m \leftarrow m - \alpha \cdot dJ/dm)$

$b \leftarrow b - \alpha \cdot dJ/db$

$m_1 = -1.45 - (0.01 \times -11.84)$

$= -1.45 + 0.1184$

$= -1.3316$

$m_2 = 0.87 - (0.01 \times -31.08)$

$= 0.87 + 0.3108$

$= 1.1808$

$b = 0.88 + 0.518 = 0.9318$

So: $m_1 = -1.3316$

$m_2 = 1.1808$

$b = 0.9318$

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Iteration 3: I2A Prince Boaz

From Iteration 2:

$m_1 = -1.3316$, $m_2 = 1.1808$

$b = 0.9318$

$\alpha = 0.01$, $n = 2$

$$X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \quad X^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$$

$$m = \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix} \quad y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

Chain rule:

$$J = \frac{1}{n} \Sigma (\hat{y} - y)^2 \quad \hat{y} = Xm + b$$

$$\frac{dJ}{dm} = \frac{dJ}{d\hat{y}} \times \frac{d\hat{y}}{dm}$$

$$= \frac{2}{n} (\hat{y} - y) \cdot X$$

$$= \frac{2}{n} X^T (\hat{y} - y)$$

$$\frac{dJ}{db} = \frac{dJ}{d\hat{y}} \cdot \frac{d\hat{y}}{db}$$

$$= \frac{2}{n} (\hat{y} - y) \cdot 1$$

$$= \frac{2}{n} \Sigma (\hat{y} - y)$$

1. Predict ($\hat{y} = mx + b$)

$$X_m = \begin{bmatrix} 1(-1.3316) + 3(1.1808) \\ 4(-1.3316) + 10(1.1808) \end{bmatrix}$$

$$= \begin{bmatrix} -1.3316 + 3.5424 \\ -5.3264 + 11.8080 \end{bmatrix}$$

$$= \begin{bmatrix} 2.2108 \\ 6.4816 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 2.2108 + 0.9318 \\ 6.4816 + 0.9318 \end{bmatrix}$$

$$= \begin{bmatrix} 3.1426 \\ 7.4134 \end{bmatrix}$$

2. Error ($e = \hat{y} - y$, $J = \frac{e_1^2 + e^2}{2}$)

$$e = \begin{bmatrix} 3.1426 - 5 \\ 7.4134 - 6 \end{bmatrix} = \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix}$$

$$J = \frac{(-1.8574)^2 + 1.4134^2}{2} = \frac{3.4499 + 1.997}{2}$$

$$J = \frac{5.4476}{2} = 2.7238$$

3. Slopes ($\frac{2}{n} = \frac{2}{2} = 1$)

$$X^T e = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix}$$

$$= \begin{bmatrix} 1(-1.8574) + 4(1.4134) \\ 3(-1.8574) + 10(1.4134) \end{bmatrix}$$

$$= \begin{bmatrix} -1.8574 + 5.6536 \\ -5.5722 + 14.1340 \end{bmatrix}$$

$$= \begin{bmatrix} 3.7962 \\ 8.5618 \end{bmatrix}$$

$$\frac{dJ}{dm} = 1 \times \begin{bmatrix} 3.7962 \\ 8.5618 \end{bmatrix} = \begin{bmatrix} 3.7962 \\ 8.5618 \end{bmatrix}$$

$$\frac{dJ}{db} = 1 \times (-1.8574 + 1.4134)$$

$$= -0.4440$$

4. Update

$$m \leftarrow m - \alpha \cdot \frac{dJ}{dm} \qquad b \leftarrow b - \alpha \cdot \frac{dJ}{db}$$

$$m_1 = -1.3316 - (0.01 \times 8.5618)$$

$$= -1.3696$$

$$m_2 = 1.1808 - (0.01 \times 8.5618)$$

$$= 1.0952$$

B/z

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$b = 0.9318 - (0.01 \times -0.4440)$

$= 0.9318 + 0.00444$

$= 0.9362$

~~After Iteration~~ 3

$w_1 = -1.3696$

$w_2 = 1.0952$

$b = 0.9362$